

Fraunhofer and Fresnel Distances

Unified derivation for aperture antennas.

In this article, the size of antenna Fresnel and Fraunhofer field regions are systematically derived starting from a general phase factor representation of the scalar diffraction theory. This is done by first expressing the phase of an arbitrary aperture field in a Taylor series in terms of a small parameter and then by subsequently imposing appropriate conditions for the Fresnel and Fraunhofer regions.

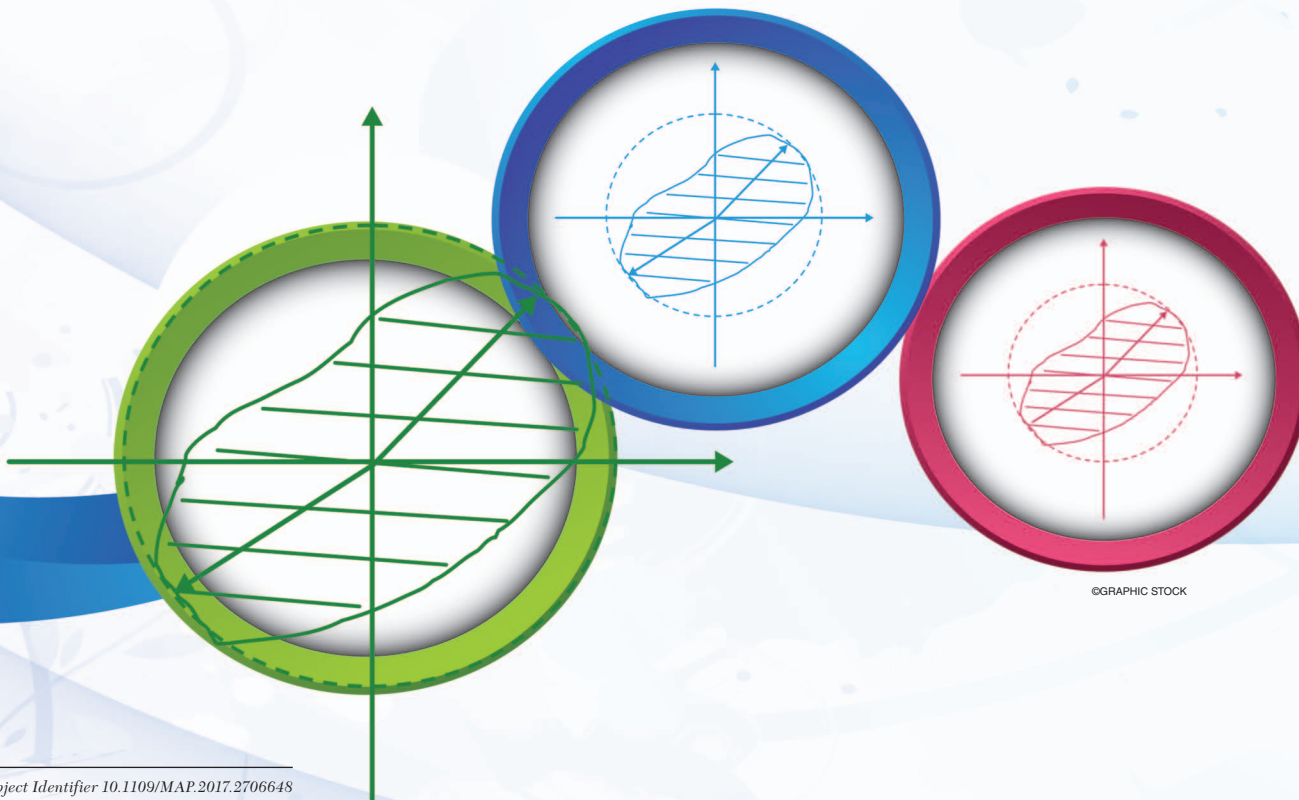
BACKGROUND

Accurate knowledge of the near- and far-field distances of an antenna are important in its characterization and in measurements. The field regions of an antenna comprise

the **Fresnel region**, in which the **angular field distribution is dependent on the distance** from the antenna, and the **Fraunhofer region**, where the **angular field distribution is independent of the distance** from the antenna. As can be inferred from many standard references (e.g., see [1, p. 32]), the Fraunhofer region can be related to the maximum allowable phase error via

$$R_f = \frac{\pi D^2}{4 \phi \lambda}, \quad (1)$$

where D is the maximum aperture dimension of the antenna, ϕ is the maximum allowable phase error, and λ is the



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wavelength. For example, (1) can be derived by assuming a simple geometry in which spherical wave fronts from an isotropic source illuminate the receiving antenna and by solving the resulting geometrical problem. For $\phi = \pi/8$, (1) reduces to the well-accepted far-field criterion $2D^2/\lambda$. The physical significance of this distance is that a radiating point source $2D^2/\lambda$ or more from a receiving antenna of length D would produce a spherical wavefront such that, over the entire length D , the phase would vary by no more than $\pi/8$ radians.

For the Fresnel boundary, at least two slightly different conditions have been reported for line sources. These are [1]–[5]:

$$R_n = 0.5 \sqrt{\frac{D^3}{\lambda}}, \quad (2)$$

and

$$R_n = 0.62 \sqrt{\frac{D^3}{\lambda}}. \quad (3)$$

As discussed in [3] and [4], (2) can be derived by considering a phase difference of $\pi/8$ between the exact ray length from the edge of a line source to the observation point and the approximate length obtained by employing the Fresnel approximations [6]. As stated in [5], (3) was first developed by Polk [6] based on scalar diffraction theory for rectangularly shaped apertures. A discussion on (3) for line sources can be found in [1]. A systematic derivation of the Fresnel region for arbitrarily shaped planar apertures is not available in the literature to the best of our knowledge, and it is the purpose of this article to fill this void. We do this by considering scalar diffraction theory for an arbitrary planar aperture, expressing the phase of the source point in terms of a Taylor series, and subsequently imposing appropriate conditions for the two regions.

THE PHASE FACTOR

Let us consider a planar aperture S , as shown in Figure 1, with a complex field distribution $\bar{F}(x', y')$. In this article, x', y' , and z' represent source coordinates in the aperture plane, and the unprimed variables (x, y, z) represent the observation point. In general, the aperture field distribution has a magnitude A and phase ψ :

$$\bar{F}(x', y') = A(x', y') e^{j\psi(x', y')}. \quad (4)$$

If we assume the aperture phase ψ to be nearly uniform, we can then write $\bar{F}$ in terms of a real-valued function A only.

With reference to Figure 1, on assuming $kr \gg 1$, we can make the following approximations, as is done in [7]:

$$\begin{aligned} \frac{1}{r} &\approx \frac{1}{R} \\ \hat{z} \cdot \hat{r} &\approx \hat{z} \cdot \hat{R} = \cos \theta. \end{aligned}$$

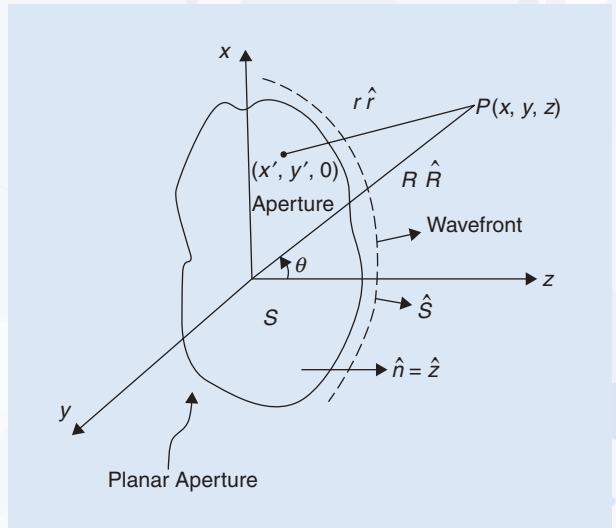


FIGURE 1. An arbitrary radiating aperture in the xy plane.

The starting point in our analysis for the field at point P is the scalar formula subject to the above approximations [7, p. 172]

$$U_P \approx \frac{j}{2\lambda R} (1 + \cos \theta) \iint_S A(x', y') e^{-jkr} dx' dy'. \quad (5)$$

From Figure 1, we have the following:

$$\begin{aligned} r^2 &= (x - x')^2 + (y - y')^2 + z^2 \\ R^2 &= x^2 + y^2 + z^2 \\ r^2 - R^2 &= x'^2 + y'^2 - 2xx' - 2yy'. \end{aligned} \quad (6)$$

The phase factor in the integrand of (5) can be written as

$$e^{-jkr} = e^{-jkR} e^{-jk(r-R)} = e^{-jkR} e^{-jk \frac{r^2 - R^2}{r + R}} = e^{-jkR} e^{-j\Phi(x, y)}, \quad (7)$$

where we have used the following representation for the phase of the source:

$$\Phi(x, y) = k \frac{r^2 - R^2}{r + R}. \quad (8)$$

By using (6), we can rewrite (8) as

$$\Phi(x', y') = k \frac{x'^2 + y'^2 - 2xx' - 2yy'}{r + R}. \quad (9)$$

With reference to Figure 2, assume that D is the smallest diameter of a circle that encloses the aperture. Then, since we would want the field boundaries to be described in terms of D , let us express the various coordinates in (9) in its terms and in terms of unitless quantities τ_x , τ_y , and ρ as follows:

$$x' = \frac{D}{2} \tau_x, y' = \frac{D}{2} \tau_y, -1 \leq \tau_x, \tau_y \leq +1, \quad (10)$$

$$R = \frac{D}{2} \rho, \rho \geq 1. \quad (11)$$

The coordinates at the observation point can be expressed similarly in a spherical coordinate system as

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$$x = \frac{D}{2}\rho\alpha, y = \frac{D}{2}\rho\beta, z = \frac{D}{2}\rho\gamma, \quad -1 \leq \alpha, \beta, \gamma \leq +1, \quad (12)$$

where α, β , and γ are the direction cosines given by $\alpha = \sin\theta \cos\phi$, $\beta = \sin\theta \sin\phi$, and $\gamma = \cos\theta$.

Using (10)–(12), the source phase in (9) can now be written as

$$\Phi(\tau_x, \tau_y) = \frac{kD}{2}(\tau_x^2 + \tau_y^2 - 2\tau_x\rho\alpha - 2\tau_y\rho\beta) \frac{1}{(\rho + \frac{2r}{D})}. \quad (13)$$

Our goal now is to make suitable approximations to the phase factor Φ in (13) so that the Fraunhofer and Fresnel distances

can be extracted from it. To this end, first consider the term appearing in the denominator of (13):

$$\frac{2r}{D} = [(\tau_x - \alpha\rho)^2 + (\tau_y - \beta\rho)^2 + (\gamma\rho)^2]^{1/2}. \quad (14)$$

We expand (14) in a Taylor series in the variables (τ_x, τ_y) up to the fourth-order, or quartic, term and use it in (13) to obtain

$$\begin{aligned} \Phi(\tau_x, \tau_y) = & -\frac{kD}{2}(\alpha\tau_x + \beta\tau_y) + \frac{kD}{4\rho}[(\tau_x^2 + \tau_y^2) - (\alpha\tau_x + \beta\tau_y)^2] \pm \\ & + \frac{kD}{8\rho^2}(\alpha\tau_x + \beta\tau_y)[(\tau_x^2 + \tau_y^2) - 2(\alpha\tau_x + \beta\tau_y)^2] \\ & - \frac{kD}{16\rho^3}(\tau_x^2 + \tau_y^2)^2 + \frac{kD}{16\rho^3}(\alpha\tau_x + \beta\tau_y)^2 \\ & \times [6(\tau_x^2 + \tau_y^2) - 5(\alpha\tau_x + \beta\tau_y)^2] \\ & + \dots \end{aligned} \quad (15)$$

Note that Φ in (15) may be viewed as a Taylor series expansion in terms of the small parameters (τ_x, τ_y) or in terms of a series in the inverse powers of the large parameter ρ .

THE FIELD BOUNDARIES

We can now approximate (15) to define boundaries for various levels of approximation to (5). For the first-order, or the Fraunhofer, region, the linear term in (15) is adequate since ρ is considered very large. The implication is that the phase error is then essentially caused by the most significant part of the ignored terms, i.e., the second-order term. **This is the phase error in the Fraunhofer region given by**

$$\delta_{FF} = \frac{kD}{4\rho}[(\tau_x^2 + \tau_y^2) - (\alpha\tau_x + \beta\tau_y)^2]. \quad (16)$$

We now investigate specifically on the beam axis where $\alpha = \beta = 0$ (recall that the aperture phase was taken to be uniform, thus resulting in a broadside beam). The maximum error is found by considering the possible maximum value of (16), subject to the constraint that $\tau_x^2 + \tau_y^2 \leq 1$. Thus,

$$\delta_{FF} \leq \frac{kD}{4\rho} = \frac{\pi}{8}. \quad (17)$$

Assuming a maximum phase error of $\delta_{FF} = \pi/8$, we arrive at the well-known result via

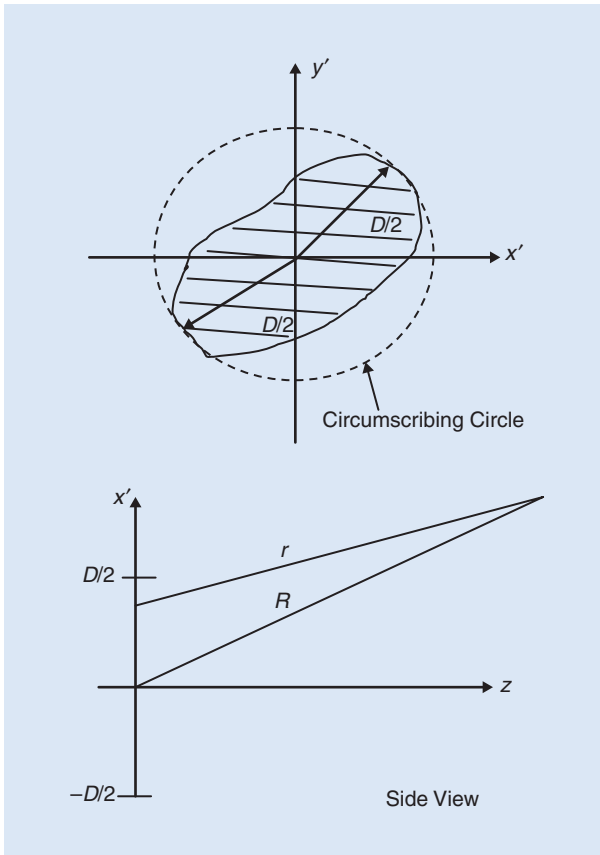


FIGURE 2. The radiating aperture enclosed within a circle of diameter D .

The implication is that the phase error is then essentially caused by the most significant part of the ignored terms.

and $\rho_{\text{FF}} = \frac{4D}{\lambda}$

$$R_{\text{FF}} = \rho_{\text{FF}} \frac{D}{2} = \frac{2D^2}{\lambda}. \quad (18)$$

We next consider the Fresnel region, for which we retain up to the second-order term in (15). The error in this approximation is due to the cubic and quartic terms in (15). When we consider the cubic term, the error caused by that term is zero on the beam axis. The quartic term then becomes the dominant source of error leading to the maximum phase error within the Fresnel boundary

$$\delta_{\text{FR}} = \frac{kD}{16\rho^3} (\tau_x^2 + \tau_y^2) \Big|_{\text{max}} = \frac{kD}{16\rho^3} = \frac{\pi D}{8\rho^3 \lambda}.$$

Setting this equal to a phase error of $\pi/8$ as done previously, we get

$$\rho_{\text{FR}}^3 = \left(\frac{2R}{D}\right)^3 = \frac{D}{\lambda}$$

and

$$R_{\text{FR}} = \left(\frac{D}{2}\right) \rho_{\text{FR}} = \frac{D}{2} \left(\frac{D}{\lambda}\right)^{1/3}. \quad (19)$$

This result for the extent of Fresnel region R_{FR} coincides with the formula given in [3, p. 30] for a line source.

Equations (18) and (19) represent the Fraunhofer and Fresnel boundaries, respectively, for the general case of an aperture antenna. Both are achieved based on a maximum phase error of $\pi/8$ radians at the demarcating boundary. In the Fraunhofer formula, it is imposed on retaining up to the linear phase term. In the Fresnel region, it is imposed on retaining up to the quadratic term for the phase.

To obtain a physical understanding of the distances involved, consider a direct broadcasting satellite receiving antenna with the diameter $D = 0.4572$ m operating at the frequency of 12.45 GHz ($\lambda = 2.41$ cm). The far-field distance (18) is equal to 1,730 m. However, the Fresnel zone distance (19) is limited to 0.61 m from the phase center of the dish antenna. This is of the order of the aperture size. Even for large antennas encountered in practice, the Fresnel distance will be of the order of the aperture. In other words, three orders of change in electrical

size will only adjust that factor by one order. A reflector antenna designed for a 3-dB beamwidth of 0.6° and a side lobe level of -30 dB will have a diameter of $D = 1.2$ m at a wavelength of 1 cm. In this case, $D/\lambda \approx 122$. Even so, $R_{\text{FR}} \approx 2.5D \approx 3$ m.

CONCLUSIONS

In this article, the well-known Fraunhofer and Fresnel bounding conditions for antennas were derived in a systematic way for an arbitrary planar aperture with uniform phase distribution. This was done by expressing the overall phase factor of the elementary sources on the aperture in terms of a Taylor series and by representing the Fraunhofer region by the linear term alone and the Fresnel region by up to the second-order terms. In each case, the phase error was then attributed to the next higher-order term(s). The desired distances were obtained by imposing a maximum error of $\pi/8$ radians.

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